

Pavan Sai Chilukala

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PROFESSIONAL SUMMARY

Full-stack and mobile engineer with production experience building and shipping web, mobile, and AI-integrated applications end to end. Skilled across React, React Native, Swift, Node.js, REST APIs, PostgreSQL, MongoDB, AWS, and CI/CD pipelines. Completed M.S. in Computer Science (GPA 3.8) in December 2025.

Product-minded engineer who thinks beyond the code — combining engineering depth with data analytics to understand business impact. Open to relocate. Available immediately.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Swift, Python, Java, C/C++, PHP, SQL
Frontend: React, React Native, Next.js, SwiftUI, Tailwind CSS, Bootstrap, HTML5/CSS3, Redux
Backend: Node.js, Express.js, Flask, REST API Design, RESTful API Integration
Databases: PostgreSQL, MySQL, MongoDB, Cassandra, Neo4j, Firebase
AI / ML: OpenAI API, Anthropic Claude API, LLM Integration, PyTorch, Machine Learning, Tableau
Cloud & DevOps: AWS S3, Google Cloud Platform, Docker, Kubernetes, CI/CD Pipelines, Vercel, Render
Tools: Git, GitHub, Postman, Xcode, WebStorm, VSCode, PyCharm, Core Audio (CATap)

PROJECTS

volumeMixer | *Swift, SwiftUI, Core Audio (CATap), macOS* | github.com/pavan-sai-ch/volumeMixer 2026

- Built a macOS menu bar utility for independent per-application volume control using low-level Core Audio Tap (CATap) APIs, allowing users to mix audio sources without affecting system-wide levels.
- Demonstrates deep Swift and macOS systems programming, working directly with Apple's audio framework at the hardware abstraction layer.

The Daily Wag | *React, PHP, MySQL, AWS S3, Docker, JWT* | github.com/pavan-sai-ch/daily-wag-frontend 2025

- Led full-stack architecture of a pet clinic platform covering grooming, vet appointments, accessories, and adoption; built decoupled system with modular component design and secure AWS S3 cloud storage.
- Engineered custom scheduling logic to prevent concurrent booking conflicts under high-traffic conditions; built admin dashboard with purchase analytics across 200+ products and loyalty-based variable pricing.

C3-Carpool Connect | *React, Flask, MongoDB Atlas, Vercel, Render, CI/CD* | github.com/pavan-sai-ch/carpool-frontend2024

- Developed a real-time ride-sharing platform matching users via optimized geo-spatial proximity queries, achieving sub-second matching latency.
- Maintained automated CI/CD pipelines on Vercel and Render, ensuring 99.9% system uptime and seamless production feature rollouts.

Black Friday Sales Prediction | *Python, Machine Learning, Data Analysis* 2022

- Developed a Python predictive ML model to analyze customer behavior patterns and optimize holiday inventory stocking decisions using historical sales data.

WORK EXPERIENCE

University of Central Missouri

Graduate Teaching Assistant — Algorithms & Data Structures

Warrensburg, MO

Jan 2025 – Dec 2025

- Supported the Design and Analysis of Algorithms course for 50+ undergraduate students, evaluating 500+ assignments for logical correctness, time complexity (Big O), and algorithmic efficiency.
- Provided structured written feedback on pseudocode, flowcharts, and implementation quality, helping students improve problem-solving approaches.
- Mentored students on selecting optimal data structures for large-scale data processing challenges, reinforcing systems thinking and scalable design.

Dhanush Infotech Pvt Ltd

React Native Developer Intern

Hyderabad, India

Sep 2022 – Dec 2022

- Built and shipped production mobile features across the full React Native stack, from UI components to API integration and App Store deployment.
- Architected 50+ modular UI components to enhance frontend scalability and maintain consistent design patterns; applied memoization techniques to achieve 80% faster turnaround across 20+ existing components.
- Integrated 100+ RESTful APIs for real-time mobile/backend synchronization, managing complex application state with Redux; improved UI responsiveness by 70% using lazy async dynamic loading.
- Published 7 production app updates to App Store and Play Store; integrated Google OAuth authentication and Firebase push notifications with 20+ deep-linked Redux navigation flows.
- Managed version control and resolved complex merge conflicts in a strict Git-flow environment, ensuring stable and compatible production releases.

EDUCATION

University of Central Missouri	Warrensburg, MO
<i>Master of Science, Computer Science GPA: 3.8 / 4.0</i>	<i>Jan 2024 – Dec 2025</i>
Jawaharlal Nehru Technological University (JNTU)	India
<i>Bachelor of Technology, Computer Science & Engineering GPA: 3.75 / 4.0</i>	<i>Aug 2019 – Jul 2023</i>

CERTIFICATIONS

React Native — The Practical Guide [2023]	Udemy
Database Design and Basic SQL in PostgreSQL	Coursera
Intermediate PostgreSQL	Coursera
Supervised Machine Learning: Regression and Classification	Coursera / DeepLearning.AI